

LSTM-Based Solar Power Prediction Using Historical and Local Data

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1 Project Description

This project is a continuation and completion of unfinished work that I originally started during an REU (Research Experiences for Undergraduates) program in the summer of 2025. The research was part of the DigiCAREs Project, which focuses on advancing renewable energy studies and smart energy systems. As part of the project, I was tasked by Dr. Fan Guiliang and Dr. Ying Zhang with designing and implementing an LSTM (Long Short-Term Memory) recurrent neural network (RNN) using Python for renewable energy prediction. Over the course of approximately five months, I developed three separate LSTM models for wind, solar, and hybrid renewable energy prediction. The primary focus of this project is the solar LSTM model, since its original performance was significantly weaker than the wind and hybrid models, producing an RMSE of approximately 1801.52. Several factors contributed to the poor performance, including inefficient preprocessing methods, poor feature selection, and limited experience with large-scale machine learning workflows at the time.

The revised version of the project focuses on redesigning and improving the original neural network pipeline by implementing cubic spline interpolation for missing-data handling, restructuring the dataset split ratios, improving validation methods, and reducing unnecessary variables through feature importance analysis. The original project used a 50/50 training and prediction split, while the revised model now uses a time-ordered 80/10/10 training, validation, and testing split to improve generalization and stability. In addition to the historical solar dataset, the project was expanded to support locally collected solar panel data from the Endeavor system provided by Evan Burk. Unlike the historical dataset, which measured large-scale solar generation in megawatts, the local dataset records minute-level measurements in watts directly from a physical solar power system. This addition allowed the project to move toward short-term local solar forecasting using both environmental and system-state variables while also supporting future live-data integration.

2 Numerical Method

The numerical method chosen to be implemented into the LSTM project was cubic spline interpolation. This method was selected because of its strong ability to handle missing values within time-series datasets, which is especially important in renewable energy prediction. In the case of solar power generation, missing or low-output values are common due to cloud coverage, poor weather conditions, sensor interruptions, or nighttime periods where solar production naturally drops to zero. Cubic spline interpolation works by estimating intermediate values between known data points using smooth cubic polynomial functions. In simpler cases, the interpolation behaves

similarly to estimating values between surrounding points while maintaining a smooth overall curve. This works well for smaller gaps in the dataset, although very large gaps can still introduce inaccuracies because significant environmental changes may occur during those missing periods.

$$S_i(x) = a_i x^3 + b_i x^2 + c_i x + d_i$$

Each interval between data points receives its own cubic polynomial while also maintaining smooth transitions between neighboring intervals through continuity in the first and second derivatives. In this project, the cubic spline interpolation method was applied before normalization and sequence generation in order to improve the overall preprocessing pipeline and reduce disruptions caused by missing data. By preserving smoother transitions within the solar generation and weather datasets, the interpolation process helped improve continuity in the time-series input data used for LSTM training. This allowed the neural network to learn patterns more effectively without needing to discard large portions of incomplete data.

3 Project Goals

Write your project goals here.

4 Model Structure

The general model structure is

Past weather/system variables + past power $\rightarrow$ future power

Write your model structure explanation here.

5 Data Splitting

The revised project uses a time-ordered split:

80% training, 10% validation, 10% testing

Write your data splitting explanation here.

6 Evaluation Metrics

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

$$SMAPE = \frac{100}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{(|y_i| + |\hat{y}_i|)/2}$$

Write your evaluation metrics explanation here.

7 Historical Dataset Results

Model	RMSE	MAE	SMAPE
Original Solar LSTM	1801.52	—	—
Persistence Baseline	714.43	411.59	57.85%
Revised Historical LSTM	659.06	291.73	56.20%

Table 1: Historical dataset model comparison.

Write your historical results explanation here.

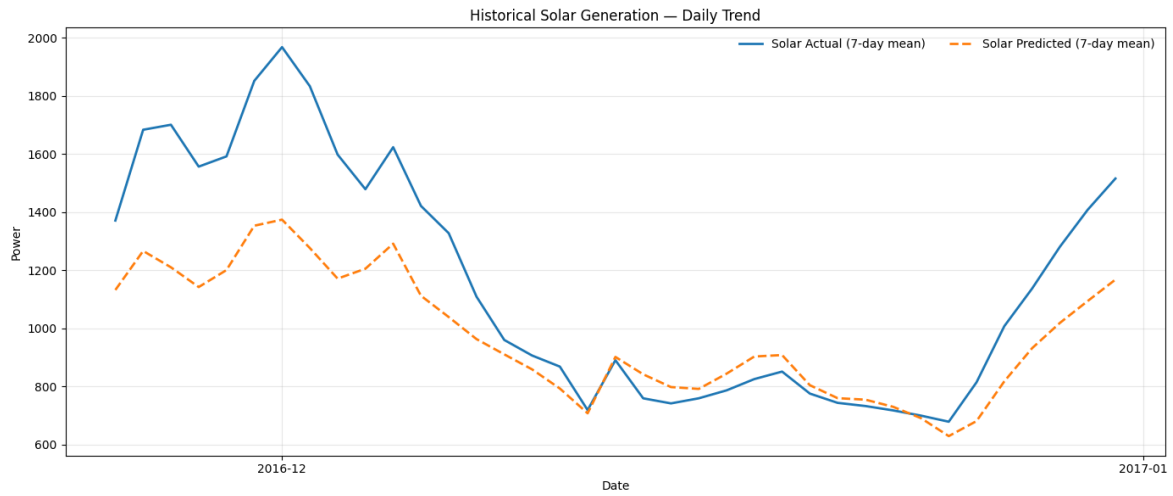


Figure 1: Historical solar generation compared against the predicted 7-day rolling mean trend.

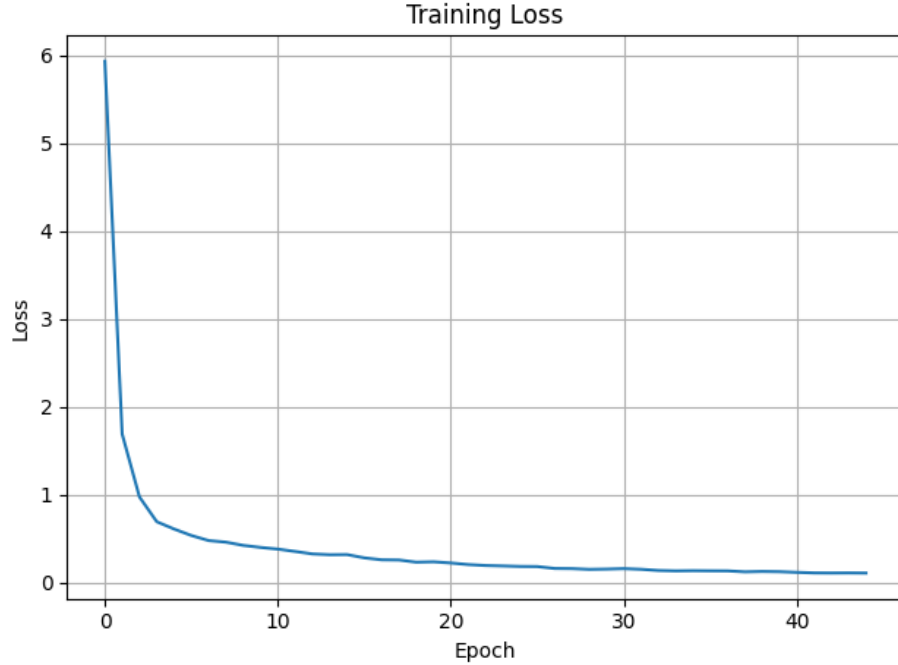


Figure 2: Training loss for the revised historical LSTM model.

8 Historical Feature Importance

Feature	RMSE Increase
SWTDN	1201.44
SWGDN	1094.27
Cloud Cover	144.90
Humidity	98.52
Dew	-47.69

Table 2: Historical permutation feature importance results.

Write your historical feature importance explanation here.

9 Local Dataset Results

Model	RMSE	MAE	SMAPE
Persistence Baseline	—	—	—
Local LSTM	7.18	—	—

Table 3: Local dataset model comparison. Fill in final persistence values from terminal output.

Write your local results explanation here.

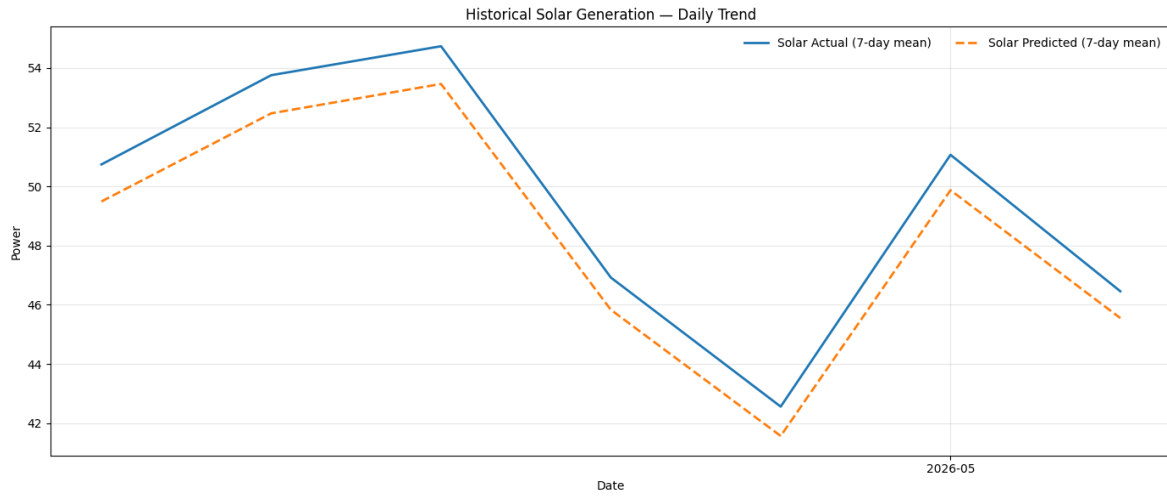


Figure 3: Local solar prediction trend using the Endeavor solar dataset.

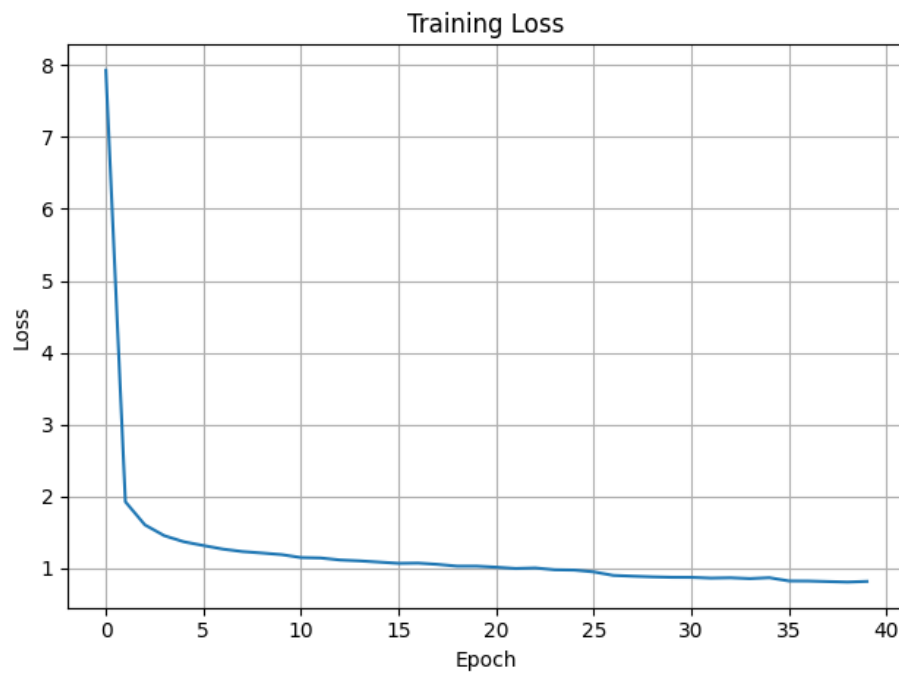


Figure 4: Training loss for the local solar prediction model.

10 Local Feature Importance

Feature	RMSE Increase
Avg Power to Battery	26.28
Battery Power	18.93
Solar Irradiance	11.14
PV Voltage	9.39
Wind Speed	0.58
PV Current	0.57
UV Index	0.37
Humidity	0.16
Air Temperature	0.13
Air Pressure	0.01

Table 4: Local dataset permutation feature importance results.

Write your local feature importance explanation here.

11 Testing and GitHub Workflow

Test File	Purpose
<code>test_basic.py</code>	Basic pytest sanity check
<code>test_metrics.py</code>	Verifies RMSE calculation
<code>test_spline_interpolation.py</code>	Verifies cubic spline missing-data handling
<code>test_model_shapes.py</code>	Verifies LSTM output dimensions

Table 5: Automated test coverage used in the project.

Write your testing and GitHub workflow explanation here.

12 Limitations and Future Work

Write your limitations and future work here.

13 Conclusion

Write your conclusion here.